

Diffraction and Interference Study Notes

Optics

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1 The Central Idea Behind Diffraction

Diffraction and interference are governed by one central chain of ideas:

$$\boxed{\text{Path difference} \longrightarrow \text{Phase difference} \longrightarrow \text{Wave addition} \longrightarrow \text{Intensity}}$$

Suppose two waves travel different distances before arriving at the same observation point. The difference between the distances travelled is called the **path difference**.

Definition 1.1. *The path difference between two waves is*

$$\Delta L = L_2 - L_1.$$

A path difference produces a phase difference because one complete wavelength corresponds to one complete phase cycle of 2π radians.

$$\Delta\phi = \frac{2\pi}{\lambda} \Delta L \quad (\text{F1.1})$$

Here:

- $\Delta\phi$ is the phase difference;
- ΔL is the path difference;
- λ is the wavelength.

Important

The geometry of an experiment gives the path difference. The path difference gives the phase difference. The phase difference determines whether waves reinforce or cancel.

1.1 Constructive and Destructive Interference

Constructive interference occurs when waves arrive in phase.

$$\Delta L = m\lambda, \quad m = 0, \pm 1, \pm 2, \dots$$

The corresponding phase difference is

$$\Delta\phi = 2m\pi.$$

Destructive interference occurs when two waves of equal amplitude arrive exactly out of phase.

$$\Delta L = \left(m + \frac{1}{2}\right) \lambda$$

and therefore

$$\Delta\phi = (2m + 1)\pi.$$

Quick Note

A path difference of λ returns a wave to the same phase. A path difference of $\lambda/2$ reverses its phase.

1.2 Why Diffraction Is an Interference Phenomenon

In an ordinary two-source interference experiment, two separate sources produce the interfering waves.

In diffraction, different portions of the *same wavefront* act as secondary sources. Waves from different parts of the opening then interfere at the observation point.

Thus,

Diffraction is interference involving continuously distributed sources.

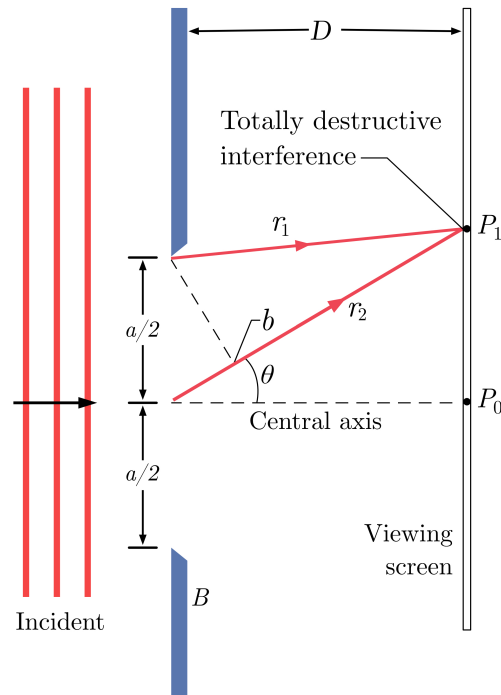
2 Single-Slit Diffraction

Consider monochromatic light of wavelength λ incident on a slit of width a .

According to the Huygens—Fresnel principle, every point across the slit behaves as a source of secondary waves.

2.1 Geometry of the Path Difference

At an observation point making an angle θ with the central axis, the ray from one edge of the slit travels farther than the ray from the opposite edge.



The total path difference between waves from the top and bottom edges is

$$\Delta L = a \sin \theta \quad (\text{F2.1})$$

where:

- a is the slit width;
- θ is the diffraction angle.

2.2 Condition for Dark Fringes

The minima of a single-slit diffraction pattern occur when

$$a \sin \theta = m\lambda, \quad m = \pm 1, \pm 2, \pm 3, \dots \quad (\text{F2.2})$$

The value $m = 0$ is excluded because

$$a \sin 0 = 0$$

corresponds to the bright central maximum, not a minimum.

Thus:

$$\begin{aligned} \text{First minimum:} & \quad a \sin \theta_1 = \lambda, \\ \text{Second minimum:} & \quad a \sin \theta_2 = 2\lambda, \\ \text{Third minimum:} & \quad a \sin \theta_3 = 3\lambda. \end{aligned}$$

2.3 Why the First Minimum Occurs

For the first minimum,

$$a \sin \theta = \lambda.$$

Divide the slit into two equal halves.

For every source point in the upper half, choose a corresponding source point in the lower half. Their separation is

$$\frac{a}{2}.$$

The path difference between the paired waves is therefore

$$\Delta L_{\text{pair}} = \frac{a}{2} \sin \theta.$$

Using the first-minimum condition,

$$a \sin \theta = \lambda,$$

we obtain

$$\Delta L_{\text{pair}} = \frac{\lambda}{2}.$$

This corresponds to a phase difference of

$$\Delta \phi = \frac{2\pi}{\lambda} \left(\frac{\lambda}{2} \right) = \pi.$$

Therefore, every contribution from one half of the slit is cancelled by a corresponding contribution from the other half.

Resultant amplitude = 0

and hence

Intensity = 0.

Personal Note

The slit is not dark because waves fail to reach the observation point. It is dark because many waves reach the point and cancel pairwise.

2.4 Positions of the Minima on a Screen

Let the screen be a distance L from the slit. Let the m -th minimum be a distance y_m from the central axis.

From the geometry,

$$\tan \theta_m = \frac{y_m}{L}.$$

For small diffraction angles,

$$\sin \theta \approx \tan \theta \approx \theta.$$

Therefore,

$$\sin \theta_m \approx \frac{y_m}{L}.$$

Using

$$a \sin \theta_m = m\lambda,$$

we obtain

$$a \left(\frac{y_m}{L} \right) = m\lambda.$$

Hence,

$$y_m = \frac{m\lambda L}{a} \tag{F2.3}$$

2.5 Width of the Central Maximum

The central maximum lies between the first minima on either side:

$$m = +1 \quad \text{and} \quad m = -1.$$

The first minimum on one side is located at

$$y_1 = \frac{\lambda L}{a}.$$

Therefore, the central maximum extends from

$$-\frac{\lambda L}{a}$$

to

$$+\frac{\lambda L}{a}.$$

Its total width is

$$W_{\text{central}} = \frac{2\lambda L}{a} \quad (\text{F2.4})$$

2.6 Physical Interpretation

From

$$W_{\text{central}} = \frac{2\lambda L}{a},$$

we see that:

$$W_{\text{central}} \propto \lambda, \quad W_{\text{central}} \propto L, \quad W_{\text{central}} \propto \frac{1}{a}.$$

Therefore:

- increasing the wavelength widens the pattern;
- increasing the screen distance widens the pattern;
- decreasing the slit width widens the pattern;
- increasing the slit width narrows the pattern.

Important

A narrower opening produces greater angular spreading. This inverse relationship between confinement and spreading is one of the deepest ideas in wave physics.

2.7 Example: Position of the First Minimum

Example 2.1. *Light of wavelength*

$$\lambda = 600 \text{ nm}$$

passes through a slit of width

$$a = 0.20 \text{ mm}.$$

The screen is

$$L = 2.0 \text{ m}$$

away. Find the position of the first minimum.

Convert the quantities to SI units:

$$\lambda = 600 \times 10^{-9} \text{ m},$$

$$a = 0.20 \times 10^{-3} \text{ m}.$$

For the first minimum, $m = 1$:

$$y_1 = \frac{\lambda L}{a}.$$

Therefore,

$$\begin{aligned} y_1 &= \frac{(600 \times 10^{-9})(2.0)}{0.20 \times 10^{-3}} \\ &= 6.0 \times 10^{-3} \text{ m}. \end{aligned}$$

Thus,

$$y_1 = 6.0 \text{ mm}.$$

The central maximum has width

$$W_{\text{central}} = 2y_1 = 12 \text{ mm}.$$

3 Intensity in Single-Slit Diffraction

The minimum condition tells us where the intensity is zero, but it does not describe the brightness at every angle.

The complete single-slit intensity distribution is

$$I(\theta) = I_{\text{max}} \left(\frac{\sin \beta}{\beta} \right)^2 \quad (\text{F3.1})$$

where

$$\beta = \frac{\pi a \sin \theta}{\lambda} \quad (\text{F3.2})$$

The function

$$\frac{\sin \beta}{\beta}$$

is related to the sinc function.

3.1 Meaning of the Parameter β

The total path difference across the slit is

$$\Delta L = a \sin \theta.$$

The corresponding total phase difference is

$$\Delta \phi = \frac{2\pi}{\lambda} \Delta L.$$

Substituting gives

$$\Delta \phi = \frac{2\pi a \sin \theta}{\lambda}.$$

The quantity β is defined as half this total phase difference:

$$\beta = \frac{\Delta \phi}{2}.$$

Therefore,

$$\boxed{\beta = \frac{\pi a \sin \theta}{\lambda}}$$

Quick Note

The parameter β is not a new physical angle in space. It is a convenient measure of the phase spread across the slit.

3.2 Origin of the Factor $\sin \beta / \beta$

Divide the slit into many very narrow strips.

Each strip contributes a small electric-field amplitude. Because the strips occupy different positions across the slit, their waves reach the observation point with slightly different phases.

These small amplitudes may be represented as phasors.

When the phasors are arranged head-to-tail:

- their total arc length represents the amplitude that would occur if all contributions were in phase;
- the chord joining the beginning and end of the arc gives the actual resultant amplitude.

The geometry of the phasor arc gives

$$E(\theta) = E_{\max} \frac{\sin \beta}{\beta}.$$

Since intensity is proportional to the square of the field amplitude,

$$I \propto E^2,$$

we obtain

$$I(\theta) = I_{\max} \left(\frac{\sin \beta}{\beta} \right)^2.$$

3.3 Intensity at the Centre

At the centre,

$$\theta = 0.$$

Therefore,

$$\beta = 0.$$

The formula appears to contain

$$\frac{\sin 0}{0}.$$

However,

$$\lim_{\beta \rightarrow 0} \frac{\sin \beta}{\beta} = 1.$$

Therefore,

$$I(0) = I_{\max}.$$

Thus, the centre is the brightest part of the diffraction pattern.

3.4 Deriving the Minima from the Intensity Formula

The intensity is zero when

$$\sin \beta = 0,$$

provided $\beta \neq 0$.

Hence,

$$\beta = m\pi, \quad m = \pm 1, \pm 2, \dots$$

Substitute

$$\beta = \frac{\pi a \sin \theta}{\lambda} :$$

$$\frac{\pi a \sin \theta}{\lambda} = m\pi.$$

Cancelling π ,

$$a \sin \theta = m\lambda.$$

Thus, the intensity equation reproduces the single-slit minimum condition.

3.5 Secondary Maxima

The secondary maxima occur between consecutive minima.

Starting from

$$I(\beta) = I_{\max} \left(\frac{\sin \beta}{\beta} \right)^2,$$

the condition for a stationary point is

$$\frac{dI}{d\beta} = 0.$$

Ignoring the minima where $\sin \beta = 0$, the secondary maxima satisfy

$$\tan \beta = \beta \quad (\text{F3.3})$$

The first few nonzero solutions are approximately

$$\beta \approx 4.493, \quad 7.725, \quad 10.904, \dots$$

The first secondary maximum has intensity approximately

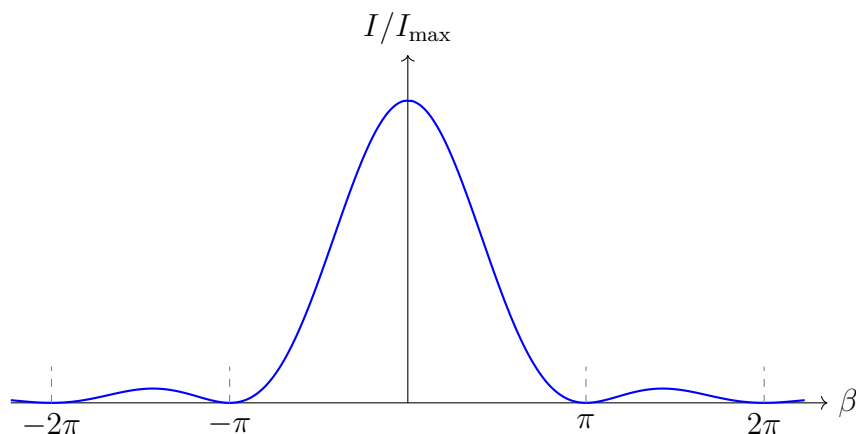
$$I \approx 0.047 I_{\max}.$$

Therefore, it is only about 4.7% as intense as the central maximum.

Important

The central maximum is both wider and much brighter than the secondary maxima.

3.6 Shape of the Intensity Pattern



4 Diffraction Through a Circular Aperture

Circular apertures occur in:

- camera lenses;
- telescope mirrors;
- microscopes;
- the human pupil;
- optical instruments.

A point source observed through a circular aperture does not form a perfect point image. Instead, it produces a bright central disc surrounded by faint rings.

This diffraction pattern is called the **Airy pattern**, and the bright central region is called the **Airy disc**.

4.1 First Minimum of a Circular Aperture

For an aperture of diameter D , the first dark ring occurs at

$$\sin \theta_{\min} = 1.22 \frac{\lambda}{D} \quad (\text{F4.1})$$

For small angles,

$$\sin \theta \approx \theta.$$

Therefore,

$$\theta_{\min} \approx 1.22 \frac{\lambda}{D} \quad (\text{F4.2})$$

The factor 1.22 arises from the circular geometry and the first zero of a Bessel function.

4.2 Exact Intensity Distribution

The exact intensity distribution is

$$I(\theta) = I_{\max} \left[\frac{2J_1(u)}{u} \right]^2 \quad (\text{F4.3})$$

where

$$u = \frac{\pi D \sin \theta}{\lambda}$$

and $J_1(u)$ is the Bessel function of the first kind of order one.

The first zero of $J_1(u)$ occurs near

$$u \approx 3.8317.$$

Hence,

$$\frac{\pi D \sin \theta}{\lambda} \approx 3.8317,$$

so

$$\sin \theta \approx \frac{3.8317}{\pi} \frac{\lambda}{D}.$$

Since

$$\frac{3.8317}{\pi} \approx 1.22,$$

we obtain

$$\sin \theta_{\min} \approx 1.22 \frac{\lambda}{D}.$$

4.3 Radius of the Airy Disc

If a screen is a distance L away, then

$$r = L \tan \theta.$$

For small angles,

$$r \approx L\theta.$$

Therefore,

$$r = 1.22 \frac{\lambda L}{D} \tag{F4.4}$$

The diameter is

$$d_{\text{Airy}} = 2.44 \frac{\lambda L}{D} \tag{F4.5}$$

If the pattern is formed in the focal plane of a lens of focal length f , replace L with f :

$$r = 1.22 \frac{\lambda f}{D}.$$

4.4 Rayleigh Criterion

Two point sources are said to be **just resolved** when the central maximum of one Airy pattern lies at the first minimum of the other.

The minimum resolvable angular separation is

$$\theta_R = 1.22 \frac{\lambda}{D} \quad (\text{F4.6})$$

Important

A smaller value of θ_R means better angular resolution.

Since

$$\theta_R \propto \frac{\lambda}{D},$$

resolution improves when:

- the aperture diameter is increased;
- the wavelength is decreased.

4.5 Example: Telescope Resolution

Example 4.1. *A telescope has an aperture diameter*

$$D = 0.20 \text{ m}$$

and observes light of wavelength

$$\lambda = 550 \text{ nm}.$$

Find its diffraction-limited angular resolution.

Use

$$\theta_R = 1.22 \frac{\lambda}{D}.$$

Thus,

$$\begin{aligned} \theta_R &= 1.22 \frac{550 \times 10^{-9}}{0.20} \\ &= 3.36 \times 10^{-6} \text{ rad}. \end{aligned}$$

Therefore,

$$\boxed{\theta_R \approx 3.36 \times 10^{-6} \text{ rad}}$$

5 Double-Slit Diffraction and Interference

For a real double-slit system, define:

- a : width of each slit;
- d : centre-to-centre slit separation;
- λ : wavelength;
- L : distance to the screen.

Two effects occur simultaneously:

Each slit diffracts	+	the two slits interfere
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5.1 Ideal Double Slit

First suppose each slit is extremely narrow, so diffraction from each slit may be neglected.

The path difference between waves from the two slit centres is

$$\Delta L = d \sin \theta \quad (\text{F5.1})$$

The corresponding phase difference is

$$\delta = \frac{2\pi}{\lambda} d \sin \theta.$$

Define

$$\alpha = \frac{\pi d \sin \theta}{\lambda} \quad (\text{F5.2})$$

so that

$$\delta = 2\alpha.$$

5.2 Deriving the Ideal Double-Slit Intensity

Suppose each slit contributes an electric-field amplitude E_0 .

The resultant amplitude of two waves with phase difference δ is

$$E = 2E_0 \cos \left(\frac{\delta}{2} \right).$$

Since

$$\frac{\delta}{2} = \alpha,$$

we have

$$E = 2E_0 \cos \alpha.$$

Intensity is proportional to the square of amplitude:

$$I \propto E^2.$$

Hence,

$$I = 4I_s \cos^2 \alpha,$$

where I_s is the intensity due to one slit.

If the maximum intensity is defined as

$$I_{\max} = 4I_s,$$

then

$$I(\theta) = I_{\max} \cos^2 \alpha \tag{F5.3}$$

or

$$I(\theta) = I_{\max} \cos^2 \left(\frac{\pi d \sin \theta}{\lambda} \right).$$

5.3 Bright Fringes

Constructive interference occurs when

$$\Delta L = m\lambda.$$

Therefore,

$$d \sin \theta = m\lambda, \quad m = 0, \pm 1, \pm 2, \dots \tag{F5.4}$$

5.4 Dark Fringes

Destructive interference occurs when

$$\Delta L = \left(m + \frac{1}{2}\right) \lambda.$$

Therefore,

$$d \sin \theta = \left(m + \frac{1}{2}\right) \lambda \quad (\text{F5.5})$$

5.5 Fringe Positions

For small angles,

$$\sin \theta \approx \frac{y}{L}.$$

Using the bright-fringe condition,

$$d \frac{y_m}{L} = m \lambda.$$

Therefore,

$$y_m = \frac{m \lambda L}{d} \quad (\text{F5.6})$$

5.6 Fringe Spacing

The spacing between consecutive bright fringes is

$$\Delta y = y_{m+1} - y_m.$$

Thus,

$$\begin{aligned} \Delta y &= \frac{(m+1)\lambda L}{d} - \frac{m\lambda L}{d} \\ &= \frac{\lambda L}{d}. \end{aligned}$$

Hence,

$$\Delta y = \frac{\lambda L}{d} \quad (\text{F5.7})$$

5.7 Real Double Slit

A real slit has finite width, so each slit produces its own diffraction pattern.

The two-slit interference fringes are therefore modulated by a single-slit diffraction envelope.

The complete intensity is

$$I(\theta) = I_0 \left(\frac{\sin \beta}{\beta} \right)^2 \cos^2 \alpha \quad (\text{F5.8})$$

where

$$\beta = \frac{\pi a \sin \theta}{\lambda}$$

and

$$\alpha = \frac{\pi d \sin \theta}{\lambda}.$$

Equivalently, if I_s is the central intensity from one slit,

$$I(\theta) = 4I_s \left(\frac{\sin \beta}{\beta} \right)^2 \cos^2 \alpha.$$

5.8 Interpretation of the Two Factors

The factor

$$\left(\frac{\sin \beta}{\beta} \right)^2$$

is the broad single-slit diffraction envelope.

The factor

$$\cos^2 \alpha$$

produces the narrow double-slit interference fringes.

Thus,

Total pattern = diffraction envelope $\times$ interference pattern
--

Personal Note

Think of the interference term as drawing narrow vertical stripes. The diffraction term acts like a large lampshade that controls how bright those stripes are.

5.9 Missing Orders

An interference maximum occurs when

$$d \sin \theta = m\lambda.$$

A diffraction minimum occurs when

$$a \sin \theta = p\lambda, \quad p = 1, 2, 3, \dots$$

A bright interference fringe disappears when both conditions occur at the same angle.

Dividing the two equations,

$$\frac{d}{a} = \frac{m}{p}.$$

Therefore,

$$m = p \frac{d}{a} \tag{F5.9}$$

Example 5.1. *Suppose*

$$d = 3a.$$

Then

$$m = 3p.$$

Hence, the missing interference orders are

$$m = 3, 6, 9, \dots$$

6 Diffraction Gratings

A diffraction grating consists of a large number of equally spaced parallel slits.

Let:

- N be the number of illuminated slits;
- d be the centre-to-centre spacing between adjacent slits;
- a be the width of each slit.

A grating behaves like a many-source interference system.

6.1 Grating Equation

The path difference between waves from adjacent slits is

$$\Delta L = d \sin \theta.$$

All adjacent waves arrive in phase when

$$d \sin \theta = m\lambda.$$

Therefore,

$$d \sin \theta = m\lambda, \quad m = 0, \pm 1, \pm 2, \dots \quad (\text{F6.1})$$

The integer m is called the **diffraction order**.

6.2 Meaning of the Orders

For $m = 0$,

$$\theta = 0,$$

which gives the central or zeroth-order maximum.

For $m = 1$,

$$d \sin \theta_1 = \lambda,$$

which gives the first-order maximum.

For $m = 2$,

$$d \sin \theta_2 = 2\lambda,$$

which gives the second-order maximum.

6.3 Grating Spacing and Line Density

Suppose a grating has n_g lines per unit length.

Then the spacing between adjacent lines is

$$d = \frac{1}{n_g} \quad (\text{F6.2})$$

Example 6.1. *A grating contains*

600 lines/mm.

Then

$$d = \frac{1}{600} \text{ mm.}$$

Since

$$1 \text{ mm} = 10^{-3} \text{ m,}$$

we obtain

$$d = \frac{10^{-3}}{600} \text{ m.}$$

Therefore,

$$\boxed{d = 1.67 \times 10^{-6} \text{ m.}}$$

6.4 Maximum Possible Order

The grating equation gives

$$\sin \theta = \frac{m\lambda}{d}.$$

Since

$$|\sin \theta| \leq 1,$$

we require

$$\frac{|m|\lambda}{d} \leq 1.$$

Thus,

$$|m| \leq \frac{d}{\lambda}.$$

The maximum possible positive order is

$$m_{\max} = \left\lfloor \frac{d}{\lambda} \right\rfloor \quad (\text{F6.3})$$

where $\lfloor x \rfloor$ means the greatest integer not exceeding x .

Quick Note

If d/λ is exactly an integer, the highest mathematical order would occur at $\theta = 90^\circ$, which may not be practically observable.

6.5 Intensity for N Narrow Slits

For N equally spaced narrow slits, define

$$\alpha = \frac{\pi d \sin \theta}{\lambda}.$$

The intensity is

$$I(\theta) = I_1 \left(\frac{\sin N\alpha}{\sin \alpha} \right)^2 \quad (\text{F6.4})$$

where I_1 is the intensity associated with one slit.

At a principal maximum,

$$\alpha = m\pi.$$

Although direct substitution gives $0/0$, the limiting value is

$$\lim_{\alpha \rightarrow m\pi} \frac{\sin N\alpha}{\sin \alpha} = N.$$

Therefore,

$$I_{\max} = N^2 I_1.$$

Thus,

$$I_{\max} \propto N^2 \quad (\text{F6.5})$$

6.6 Why Many Slits Produce Sharp Peaks

Suppose the phase difference between adjacent slits changes slightly by an amount ε .

The phase difference between the first and last slit is approximately

$$(N - 1)\varepsilon.$$

When N is large, even a small ε produces a large total phase spread.

Consequently, the waves cancel strongly except very close to the principal maxima.

Therefore, increasing N :

- makes the principal maxima brighter;
- makes them narrower;
- improves the ability to distinguish nearby wavelengths.

6.7 Finite Slit Width

If each grating slit has width a , the full intensity becomes

$$I(\theta) = I_1 \left(\frac{\sin \beta}{\beta} \right)^2 \left(\frac{\sin N\alpha}{\sin \alpha} \right)^2 \quad (\text{F6.6})$$

where

$$\beta = \frac{\pi a \sin \theta}{\lambda}$$

and

$$\alpha = \frac{\pi d \sin \theta}{\lambda}.$$

The factor

$$\left(\frac{\sin \beta}{\beta} \right)^2$$

is the single-slit diffraction envelope.

The factor

$$\left(\frac{\sin N\alpha}{\sin \alpha} \right)^2$$

is the many-slit interference pattern.

6.8 Angular Width of a Principal Maximum

The approximate angular width between the neighbouring minima around the m -th principal maximum is

$$\Delta\theta \approx \frac{2\lambda}{Nd \cos \theta_m} \quad (\text{F6.7})$$

The half-width is

$$\delta\theta \approx \frac{\lambda}{Nd \cos \theta_m}.$$

Therefore,

$$\Delta\theta \propto \frac{1}{N}.$$

More illuminated slits produce narrower maxima.

6.9 Angular Dispersion

The grating equation is

$$d \sin \theta = m\lambda.$$

Differentiate with respect to λ , treating d and m as constants:

$$d \cos \theta \frac{d\theta}{d\lambda} = m.$$

Therefore,

$$\frac{d\theta}{d\lambda} = \frac{m}{d \cos \theta} \quad (\text{F6.8})$$

Angular dispersion increases when:

- the order m increases;
- the grating spacing d decreases;
- the diffraction angle increases.

6.10 Resolving Power

The resolving power of a grating is defined as

$$R = \frac{\lambda}{\Delta\lambda},$$

where $\Delta\lambda$ is the smallest wavelength difference that can be resolved.

For a diffraction grating,

$$R = \frac{\lambda}{\Delta\lambda} = mN \quad (\text{F6.9})$$

Therefore,

$$\Delta\lambda_{\min} = \frac{\lambda}{mN} \quad (\text{F6.10})$$

Resolution improves when:

- a higher diffraction order is used;
- more grating lines are illuminated.

6.11 Example: Grating Angle

Example 6.2. *Light of wavelength*

$$\lambda = 500 \text{ nm}$$

is incident on a grating with

$$600 \text{ lines/mm.}$$

Find the first-order diffraction angle.

The grating spacing is

$$d = \frac{10^{-3}}{600} = 1.67 \times 10^{-6} \text{ m.}$$

For the first order, $m = 1$:

$$d \sin \theta = \lambda.$$

Therefore,

$$\sin \theta = \frac{\lambda}{d}.$$

Substituting,

$$\sin \theta = \frac{500 \times 10^{-9}}{1.67 \times 10^{-6}} \approx 0.300.$$

Hence,

$$\boxed{\theta \approx 17.5^\circ.}$$

7 X-Ray Diffraction

Visible light has wavelengths of approximately

$$\lambda_{\text{visible}} \sim 400\text{--}700 \text{ nm}.$$

Typical atomic spacings are approximately

$$d \sim 0.1 \text{ nm}.$$

Visible light therefore has a wavelength much larger than the spacing between atoms.

X-rays, however, commonly have wavelengths of order

$$\lambda_{\text{X-ray}} \sim 10^{-10} \text{ m},$$

which is comparable to atomic spacings.

A crystal can therefore act as a three-dimensional diffraction grating for X-rays.

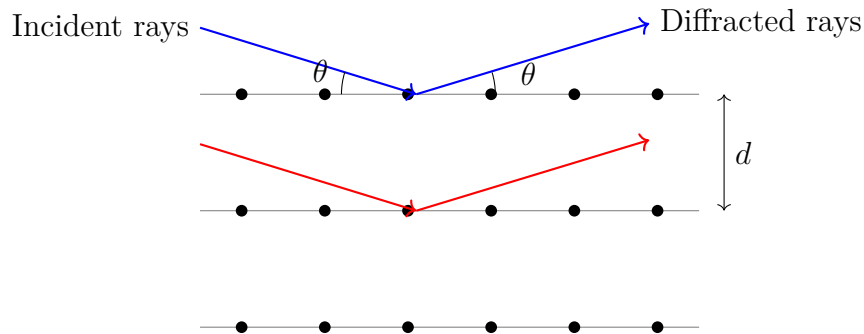
7.1 Crystal Planes

Atoms in a crystal may be regarded as lying in sets of parallel planes separated by a distance d .

X-rays incident on these planes are scattered by the atoms.

Strong reflected intensity occurs when the waves scattered from successive planes interfere constructively.

7.2 Geometry of Bragg Diffraction



One ray reflects from an upper crystal plane. Another penetrates to the next plane before being reflected.

The second ray travels extra distance on the way into the crystal and again on the way out.

Each extra segment has length

$$d \sin \theta.$$

Therefore, the total path difference is

$$\Delta L = d \sin \theta + d \sin \theta.$$

Thus,

$$\Delta L = 2d \sin \theta \quad (\text{F7.1})$$

7.3 Bragg's Law

Constructive interference occurs when the path difference is an integer multiple of the wavelength:

$$2d \sin \theta = n\lambda.$$

Therefore,

$$n\lambda = 2d \sin \theta \quad (\text{F7.2})$$

where:

- $n = 1, 2, 3, \dots$ is the diffraction order;
- λ is the X-ray wavelength;

- d is the spacing between crystal planes;
- θ is the Bragg angle.

7.4 Why the Factor of Two Appears

The deeper ray travels an additional distance twice:

$$\text{extra distance entering} = d \sin \theta,$$

and

$$\text{extra distance leaving} = d \sin \theta.$$

Therefore,

$$\Delta L = d \sin \theta + d \sin \theta = 2d \sin \theta.$$

7.5 The 2θ Convention

In many X-ray diffraction instruments, the angle displayed by the detector is the angle between the incident and diffracted beams.

This angle is

$$2\theta.$$

However, Bragg's law uses θ :

$$n\lambda = 2d \sin \theta.$$

Important

If an X-ray diffraction problem gives the measured scattering angle as 2θ , divide it by two before substituting into Bragg's law.

7.6 Example: Finding Crystal-Plane Spacing

Example 7.1. *X-rays of wavelength*

$$\lambda = 0.154 \text{ nm}$$

produce a first-order Bragg peak at

$$\theta = 30^\circ.$$

Find the spacing between the crystal planes.

For first order,

$$n = 1.$$

Bragg's law gives

$$n\lambda = 2d \sin \theta.$$

Therefore,

$$d = \frac{n\lambda}{2 \sin \theta}.$$

Substituting,

$$d = \frac{0.154}{2 \sin 30^\circ} \text{ nm.}$$

Since

$$\sin 30^\circ = \frac{1}{2},$$

we obtain

$$d = \frac{0.154}{1} \text{ nm.}$$

Thus,

$$\boxed{d = 0.154 \text{ nm.}}$$

7.7 Uses of X-Ray Diffraction

X-ray diffraction can be used to determine:

- crystal-plane spacings;
- crystal structures;
- lattice constants;

- material phases;
- molecular arrangements;
- strain and defects in crystalline materials.

8 Comparison of the Main Formulas

Arrangement	Important dimension	Main formula	What it locates
Single slit	Slit width a	$a \sin \theta = m\lambda$	Dark minima
Double slit	Slit separation d	$d \sin \theta = m\lambda$	Bright fringes
Diffraction grating	Grating spacing d	$d \sin \theta = m\lambda$	Principal maxima
Circular aperture	Diameter D	$\theta_{\min} = 1.22\lambda/D$	First dark ring
Crystal planes	Plane spacing d	$n\lambda = 2d \sin \theta$	X-ray diffraction maxima

8.1 The Most Important Distinction

For a single slit,

$$a \sin \theta = m\lambda \quad \text{locates minima.}$$

For a double slit or diffraction grating,

$$d \sin \theta = m\lambda \quad \text{locates maxima.}$$

The equations look similar, but they describe different features of the patterns.

Single slit: integer wavelengths give darkness.

Double slit and grating: integer wavelengths give brightness.

Remember:
S

9 Problem-Solving Strategy

When solving a diffraction problem, use the following method.

9.1 Step 1: Identify the Arrangement

Ask whether the problem involves:

- one slit;
- two slits;
- many slits;
- a circular aperture;
- crystal planes.

9.2 Step 2: Identify the Relevant Dimension

Determine whether the important dimension is:

- slit width a ;
- slit separation d ;
- aperture diameter D ;
- crystal-plane spacing d .

9.3 Step 3: Identify What Is Being Requested

Determine whether the problem asks for:

- a minimum;
- a maximum;
- fringe position;
- fringe spacing;
- central-maximum width;
- angular resolution;
- grating order;
- crystal spacing;
- intensity.

9.4 Step 4: Choose the Correct Equation

Use:

$$a \sin \theta = m\lambda$$

for single-slit minima;

$$d \sin \theta = m\lambda$$

for double-slit or grating maxima;

$$\theta_R = 1.22 \frac{\lambda}{D}$$

for circular-aperture resolution;

$$n\lambda = 2d \sin \theta$$

for X-ray diffraction.

9.5 Step 5: Check the Angle Convention

For screen-position problems, use

$$\tan \theta = \frac{y}{L}.$$

For small angles,

$$\sin \theta \approx \tan \theta \approx \frac{y}{L}.$$

For X-ray diffraction, check whether the stated angle is θ or 2θ .

9.6 Step 6: Convert All Units

Common conversions include

$$1 \text{ nm} = 10^{-9} \text{ m},$$

$$1 \mu\text{m} = 10^{-6} \text{ m},$$

$$1 \text{ mm} = 10^{-3} \text{ m}.$$

9.7 Step 7: Check Physical Possibility

For grating problems,

$$|\sin \theta| \leq 1.$$

Therefore, a calculated value such as

$$\sin \theta = 1.4$$

means that the proposed diffraction order does not exist.

10 Common Mistakes

1. Using the single-slit minimum formula as though it located bright fringes.
2. Confusing slit width a with slit separation d .
3. Forgetting that a real double slit contains both diffraction and interference.
4. Using line density directly as the grating spacing instead of taking its reciprocal.
5. Forgetting to convert lines per millimetre into lines per metre.
6. Using the full measured X-ray angle 2θ in Bragg's law.
7. Using small-angle approximations when the angle is not small.
8. Including $m = 0$ as a single-slit minimum.
9. Forgetting that intensity is proportional to the square of amplitude.
10. Assuming that narrower slits produce narrower diffraction patterns.

11 Formula Sheet

11.1 General Phase Relation

$$\Delta\phi = \frac{2\pi}{\lambda} \Delta L$$

11.2 Single Slit

$$a \sin \theta = m\lambda \quad \text{minima}$$

$$y_m = \frac{m\lambda L}{a}$$

$$W_{\text{central}} = \frac{2\lambda L}{a}$$

$$I(\theta) = I_{\text{max}} \left(\frac{\sin \beta}{\beta} \right)^2$$

$$\beta = \frac{\pi a \sin \theta}{\lambda}$$

$$\tan \beta = \beta \quad \text{secondary maxima}$$

11.3 Circular Aperture

$$\theta_{\text{min}} = 1.22 \frac{\lambda}{D}$$

$$r = 1.22 \frac{\lambda L}{D}$$

$$d_{\text{Airy}} = 2.44 \frac{\lambda L}{D}$$

$$\theta_R = 1.22 \frac{\lambda}{D}$$

11.4 Double Slit

$$d \sin \theta = m\lambda \quad \text{bright fringes}$$

$$d \sin \theta = \left(m + \frac{1}{2} \right) \lambda \quad \text{dark fringes}$$

$$y_m = \frac{m\lambda L}{d}$$

$$\Delta y = \frac{\lambda L}{d}$$

$$I = I_{\max} \cos^2 \alpha$$

$$\alpha = \frac{\pi d \sin \theta}{\lambda}$$

$$I = I_0 \left(\frac{\sin \beta}{\beta} \right)^2 \cos^2 \alpha$$

11.5 Diffraction Grating

$$d \sin \theta = m\lambda$$

$$d = \frac{1}{n_g}$$

$$m_{\max} = \left\lfloor \frac{d}{\lambda} \right\rfloor$$

$$I = I_1 \left(\frac{\sin N\alpha}{\sin \alpha} \right)^2$$

$$\Delta \theta \approx \frac{2\lambda}{Nd \cos \theta_m}$$

$$\frac{d\theta}{d\lambda} = \frac{m}{d \cos \theta}$$

$$R = \frac{\lambda}{\Delta \lambda} = mN$$

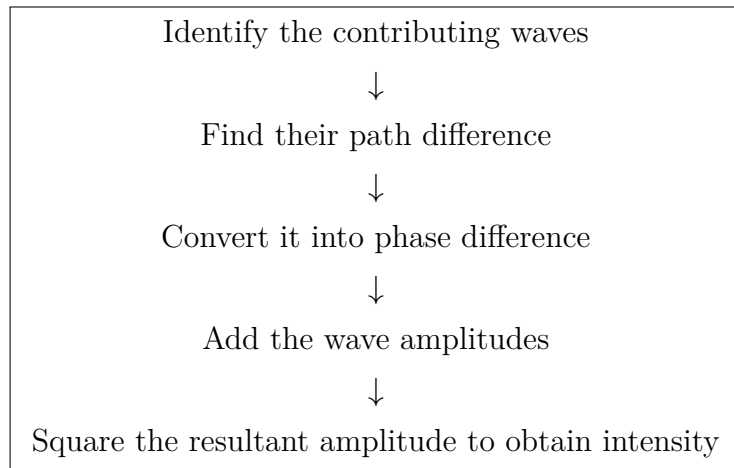
11.6 X-Ray Diffraction

$$\Delta L = 2d \sin \theta$$

$$n\lambda = 2d \sin \theta$$

12 Final Conceptual Map

Every diffraction formula follows the same reasoning:



The physical systems differ, but the logic remains the same:

Geometry determines phase, and phase determines brightness.